

# A Boundary-Orbit Far Atom and a Signed Diffuse-Compensation Obstruction

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## Abstract

The corrected RH-334 observation map defines an actual physical signed far slot at the first alias. We isolate within that slot a real deterministic boundary-orbit atom. For the primitive period- $2k$  boundary cycle, all but one of its folded marked points lie in the frozen far set for every fixed window parameter and all sufficiently small noise. The resulting set has  $2k - 1$  points. Its noisy localized trace is exactly zero because a finite set has zero multiplication operator on  $L^2$ , whereas its corrected deterministic mass is

$$D_k^{\text{orb}} = r_H^{-2k} \frac{2k - 1}{1 + |M_k|}.$$

Thus its signed far subledger is  $-D_k^{\text{orb}}$ . The physical multiplier law gives  $D_k^{\text{orb}}/\mathcal{A}_{k,2k} \rightarrow 1$  and  $D_k^{\text{orb}}/H_k \rightarrow +\infty$ . This is an alias-sized physical obstruction to proving the far estimate by taking separate absolute values after the orbit/rest split. It is not an aggregate far nonvanishing theorem. If the full far slot is  $o(H_k)$ , the complementary diffuse ledger must compensate the orbit atom as  $D_k^{\text{orb}} + o(H_k)$ , requiring relative precision  $o((\beta R)^{-2k})$ . No moving-order estimate for that diffuse term is available, so aggregate far closure and aggregate nonvanishing both remain not testable. No determinant or Riemann-hypothesis conclusion follows.

## 1 Frozen far observation and boundary orbit

Let  $u_c \in (1, 2)$  solve

$$u_c^3 - 2u_c^2 + 2u_c - 2 = 0 \tag{1}$$

and put

$$f(x) = 1 - u_c x^2, \quad r = u_c - 1, \quad \lambda = 2u_c r > 1, \quad b = u_c^{-1/2}. \tag{2}$$

Then  $f(r) = r$ ,  $f(1) = -r$ , and the two-step component map is  $S = f^2$ . The inverse branch used by the primitive boundary word is

$$h(x) = \sqrt{\frac{1 - \sqrt{(1-x)/u_c}}{u_c}}, \quad h(1) = b, \tag{3}$$

with  $h$  increasing and  $h(x) < x$  on  $(r, 1)$  [1].

Fix  $A > 0$ . RH-334 freezes the boundary-owned windows before evaluating any trace:

$$J_{\sigma,k,A}^- = [0, 1] \cap [b - A\sqrt{\sigma}, b), \tag{4}$$

$$J_{\sigma,k,A}^+ = [0, 1] \cap [b, b + A\sqrt{\sigma}], \tag{5}$$

$$F_{\sigma,k,A} = [0, 1] \setminus (J_{\sigma,k,A}^- \cup J_{\sigma,k,A}^+). \tag{6}$$

For measurable  $J \subset [0, 1]$  and  $n \geq 2$ , let

$$L_{\sigma,n}(J) = \text{Tr}(M_J K_\sigma^n), \quad (7)$$

$$P_n^{\text{abs}}(J) = \sum_{\substack{f^n(x)=x \\ |x| \in J}} \frac{1}{|1 - (f^n)'(x)|}. \quad (8)$$

Here  $K_\sigma$  is the RH-334 folded backward-observable noisy operator; the multiplication operator marks the cyclic basepoint rather than a forward probability event. The corrected actual far slot is

$$\mathcal{R}_k := r_H^{-2k} \{L_{\sigma,2k}(F_{\sigma,k,A}) - P_{2k}^{\text{abs}}(F_{\sigma,k,A})\}, \quad r_H = \frac{17}{20}. \quad (9)$$

Absolute-value folding is a multiplier-preserving bijection between the signed and folded fixed points, including marked-point multiplicity and minimal period [6].

Let  $p_k = p_{2k}$  be the primitive boundary point coded by  $CA(CB)^{k-1}$ . RH-17 proves that its  $S$ -orbit is the exact ordered chain

$$p_k, h^{k-1}(p_k), h^{k-2}(p_k), \dots, h(p_k), \quad (10)$$

and that

$$M_k := (f^{2k})'(p_k) = (S^k)'(p_k) = -C_M \lambda^k \{1 + o(1)\}, \quad C_M > 0. \quad (11)$$

The word is primitive, so all  $2k$  signed and folded marked points are distinct [1, 5].

We work on the exact physical natural clock

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \sigma \rightarrow 0, \quad (12)$$

though the containment argument below needs only  $k \rightarrow \infty$  and  $\sigma \rightarrow 0$ .

## 2 The physical far-orbit atom

For the signed orbit  $x_j = f^j(p_k)$ , define the folded marked set with one point deleted:

$$\Omega_k = \{|x_j| : 0 \leq j < 2k, j \neq 2k-2\}. \quad (13)$$

The deleted point is the final even component point  $|x_{2k-2}| = h(p_k)$ . Its distance from  $b$  is itself on the critical  $\lambda^{-k}$  scale, so no fixed- $A$  membership is asserted for it.

**Theorem 2.1** (Eventual far containment of  $2k-1$  marked points). *For every fixed  $A > 0$ , along (12) and all sufficiently large  $k$ ,*

$$\Omega_k \subset F_{\sigma,k,A}, \quad |\Omega_k| = 2k-1. \quad (14)$$

*Proof.* RH-17 proves that  $p_k$  increases with  $k$  and

$$p_k \geq p_1 > b. \quad (15)$$

Thus the endpoint  $p_k$  lies to the right of the shrinking windows by the fixed gap  $p_1 - b > 0$ .

The even marked points are the  $S$ -chain (10). Except for  $h(p_k)$ , every internal even point has the form  $h^m(p_k)$  with  $m \geq 2$ . Monotonicity and  $p_k < 1$  give

$$h^m(p_k) \leq h^2(1) = h(b) < b. \quad (16)$$

Hence these points lie to the left by the fixed gap  $b - h(b) > 0$ .

Every point in the positive  $S$ -chain lies in  $[r, 1]$ . Since  $f$  decreases from  $f(r) = r$  to  $f(1) = -r$  on this interval, every odd folded point satisfies

$$|f(S^j(p_k))| \leq r < b, \quad (17)$$

with fixed gap  $b - r > 0$ . Choose  $k$  large enough that

$$A\sqrt{\sigma} < \min\{p_1 - b, b - h(b), b - r\}.$$

Equations (15)–(17) then place every marked point except  $h(p_k)$  in  $F_{\sigma,k,A}$ . Primitivity and the folding bijection give distinctness, so deleting one of  $2k$  points leaves cardinality  $2k - 1$ .  $\square$

**Theorem 2.2** (Exact signed far-orbit subledger). *For all  $k$  in the range of theorem 2.1, define*

$$D_k^{\text{orb}} = r_H^{-2k} \frac{2k - 1}{1 + |M_k|} > 0. \quad (18)$$

Then

$$\boxed{\mathcal{R}_k^{\text{orb}} := r_H^{-2k} \{L_{\sigma,2k}(\Omega_k) - P_{2k}^{\text{abs}}(\Omega_k)\} = -D_k^{\text{orb}}.} \quad (19)$$

Moreover the full actual far slot has the exact decomposition

$$\mathcal{R}_k = \mathcal{R}_k^{\text{orb}} + \mathcal{R}_k^{\text{rest}}, \quad (20)$$

where  $\mathcal{R}_k^{\text{rest}}$  is the same localized ledger on  $F_{\sigma,k,A} \setminus \Omega_k$ .

*Proof.* A finite set has indicator zero almost everywhere, so  $M_{\Omega_k} = 0$  as an operator on  $L^2[0, 1]$ . Therefore  $L_{\sigma,2k}(\Omega_k) = 0$  exactly. The derivative of  $f^{2k}$  is the same cyclic product at every marked point of one orbit. By (11) it is negative for large  $k$ , and the RH-334 folding theorem preserves it. Each of the  $2k - 1$  folded marked points therefore has deterministic weight

$$\frac{1}{|1 - M_k|} = \frac{1}{1 + |M_k|}.$$

This proves (19). Additivity of the localized noisy trace and of the deterministic fixed-point sum across the disjoint partition  $F = \Omega_k \sqcup (F \setminus \Omega_k)$  proves (20).  $\square$

*Remark 2.3* (The windows remain the frozen observation).  $\Omega_k$  is an analytic subpartition of the already-frozen far set; it is not promoted to a new canonical physical window or a fitted observation coefficient. Its dependence on the known deterministic orbit is used only to expose one exact signed constituent of  $\mathcal{R}_k$ . The aggregate far slot and all claims about it retain the original RH-334 windows.

### 3 Alias scale and signed compensation

Retain

$$\beta = \frac{1}{r_H \sqrt{\lambda}}, \quad H_k = kR^{-2k}, \quad R = \frac{7}{5}. \quad (21)$$

The physical first-alias packet of RH-326 [4] is

$$\mathcal{A}_{k,2k} = (2k - 2)\beta_k^{2k} + 2\beta_k^{2k}, \quad \beta_k^{2k} = \frac{r_H^{-2k}}{|M_k|}. \quad (22)$$

**Theorem 3.1** (The far atom is alias-sized and super-target). *As  $k \rightarrow \infty$ ,*

$$D_k^{\text{orb}} = \frac{2k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad (23)$$

$$\frac{D_k^{\text{orb}}}{\mathcal{A}_{k,2k}} \rightarrow 1, \quad (24)$$

$$\frac{D_k^{\text{orb}}}{H_k} = \frac{2}{C_M} (\beta R)^{2k} \{1 + o(1)\} \rightarrow +\infty. \quad (25)$$

*Proof.* Insert (11) into (18) and use  $\beta^{2k} = r_H^{-2k} \lambda^{-k}$  to obtain (23). The same multiplier law in (22) gives

$$\mathcal{A}_{k,2k} = \frac{2k}{C_M} \beta^{2k} \{1 + o(1)\},$$

which proves (24). Division by  $H_k$  gives the first identity in (25). RH-336 certifies  $\beta R > 1$  exactly, without relying on a rounded decimal [7]; hence the limit is infinite.  $\square$

**Corollary 3.2** (Necessary diffuse compensation). *The aggregate far estimate  $\mathcal{R}_k = o(H_k)$  holds if and only if*

$$\mathcal{R}_k^{\text{rest}} = D_k^{\text{orb}} + o(H_k). \quad (26)$$

*Relative to the orbit mass, the required precision is*

$$o\left(\frac{H_k}{D_k^{\text{orb}}}\right) = o\left((\beta R)^{-2k}\right). \quad (27)$$

*In particular, taking separate absolute values after the atom/rest split cannot prove  $\mathcal{R}_k = o(H_k)$ , because*

$$\frac{|\mathcal{R}_k^{\text{orb}}| + |\mathcal{R}_k^{\text{rest}}|}{H_k} \geq \frac{D_k^{\text{orb}}}{H_k} \rightarrow +\infty. \quad (28)$$

*Proof.* Substitute  $\mathcal{R}_k^{\text{orb}} = -D_k^{\text{orb}}$  into (20). This proves the equivalence. The reciprocal of (25) gives (27), up to a harmless positive constant. Inequality (28) is immediate.  $\square$

This corollary does not say that signed compensation is absent. It says precisely what an aggregate positive theorem must prove. RH-19's local tail analysis explicitly leaves genuine bulk excursions to a missing resolvent or moving-order theorem [2]; the repository contains no estimate of  $\mathcal{R}_k^{\text{rest}}$  at the scale in (27). Accordingly both  $\mathcal{R}_k = o(H_k)$  and aggregate nonvanishing remain NOT\_TESTABLE. After the clock obstruction of RH-337 there is also no valid hatted actual/model pair typing a physical  $\Delta \mathcal{R}_k$  [3]. The result here concerns the actual physical far slot itself, not RH-330 transfer [8].

## 4 Reproduction and claim boundary

The artifact solves the physical algebraic constants and boundary inverse word at high Decimal precision. For  $k = 2, 4, 8, 16, 32$  and the diagnostic choice  $A = 1/4$ , it checks the counts 3, 7, 15, 31, 63, orbit closure, multiplier sign, approach of  $D_k^{\text{orb}}/\mathcal{A}_{k,2k}$  to one, and growth of  $D_k^{\text{orb}}/H_k$ . These rows reproduce the analytic formulas; they are not interval certificates and do not prove the moving theorem.

The maximum rigorous conclusion is the exact physical orbit subledger, its alias-scale asymptotic, and the necessary signed compensation law. The paper does not bound the diffuse rest, prove aggregate far vanishing or nonvanishing, close the parity–alias or off–alias packets, transport the noisy head, glue a determinant, or prove full–trace divergence. No Hilbert–Polya operator, Riemann-zero identification, von Mangoldt trace, completed-zeta divisor equality, or Riemann-hypothesis proof is obtained. Gates A–E remain false/open.

## References

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